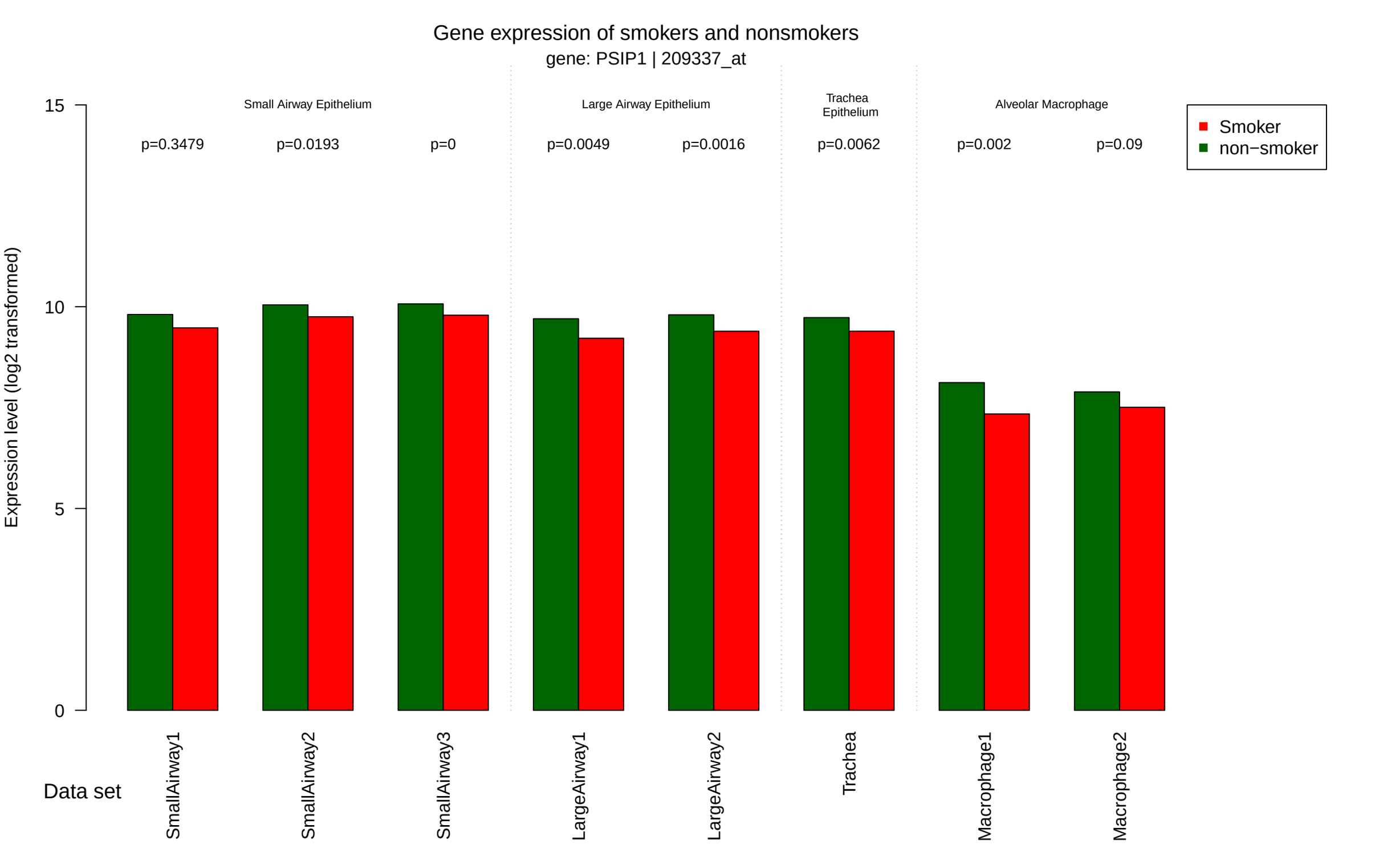
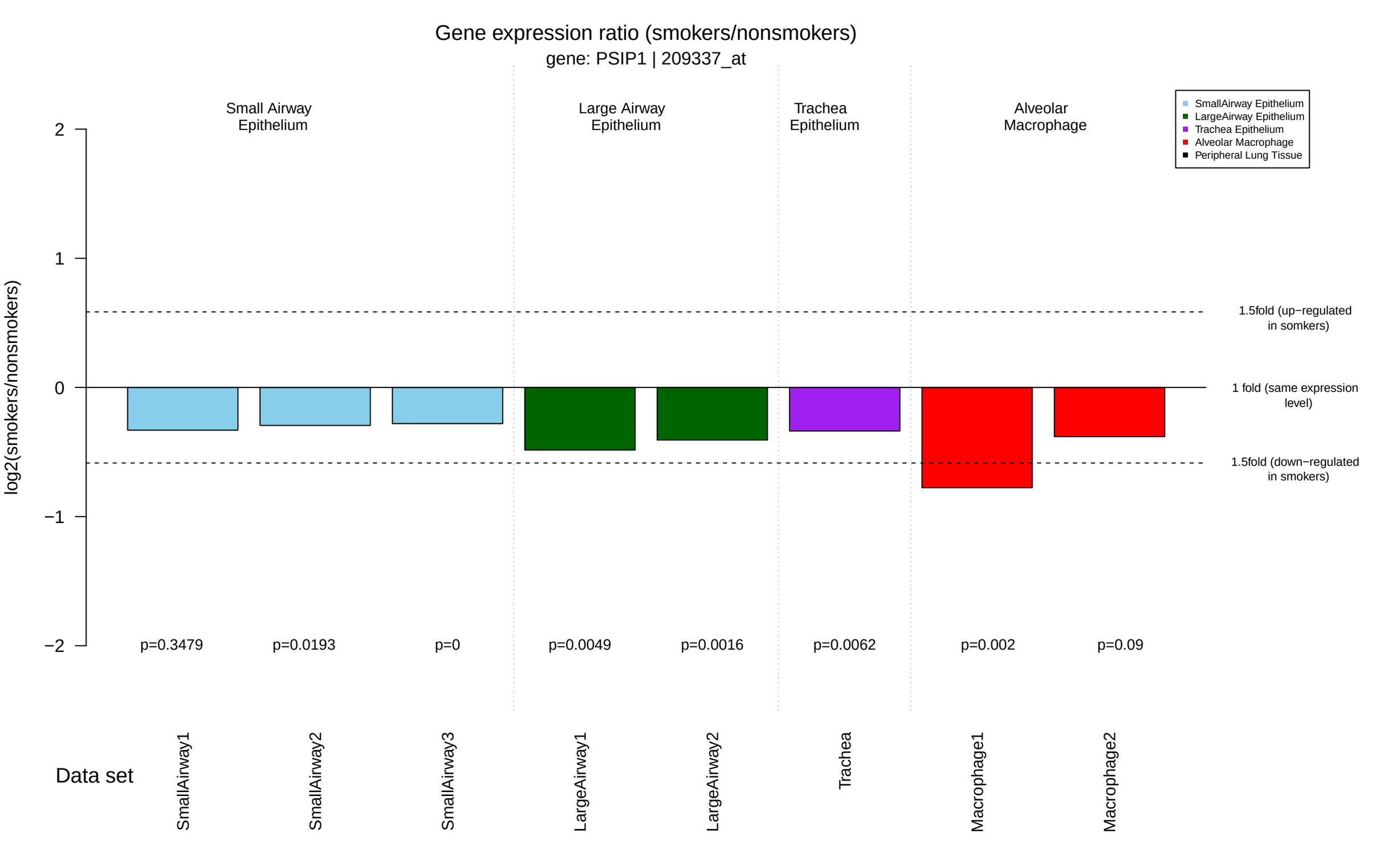
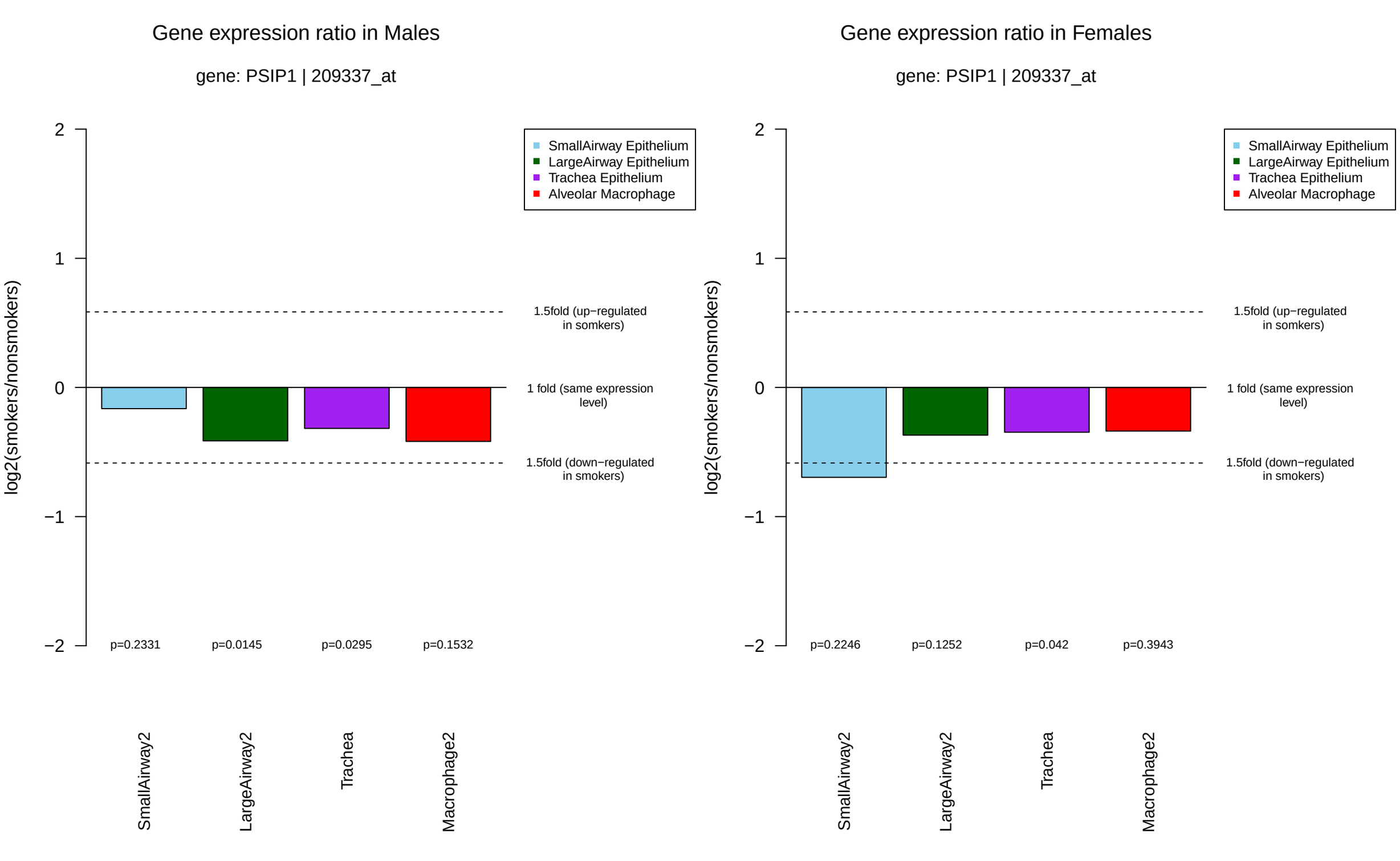
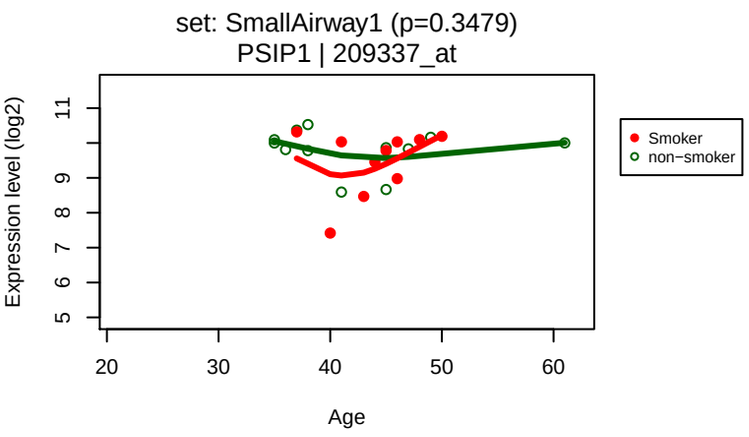
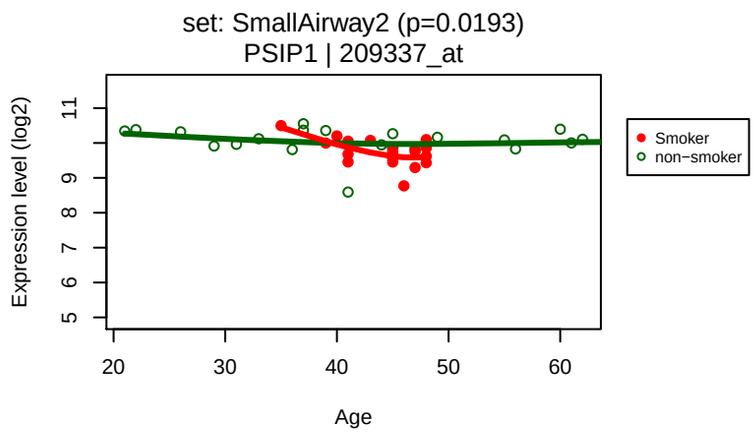
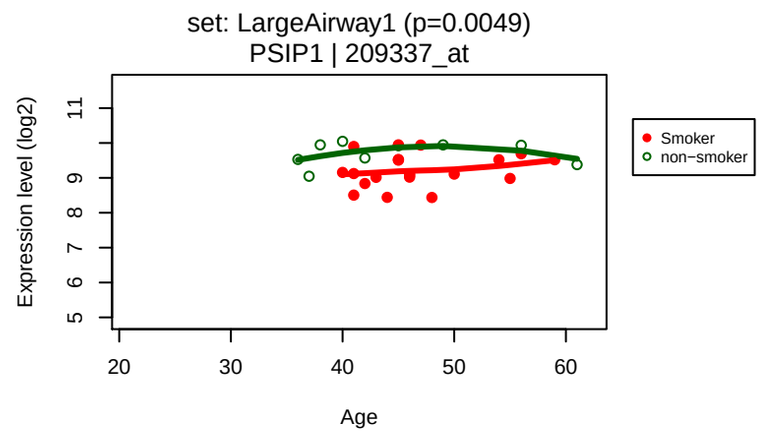

Gene expression of smokers and nonsmokers
gene: PSIP1 | 209337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3479
p=0.0193
p=0
p=0.0049
p=0.0016
p=0.0062
p=0.002
p=0.09
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

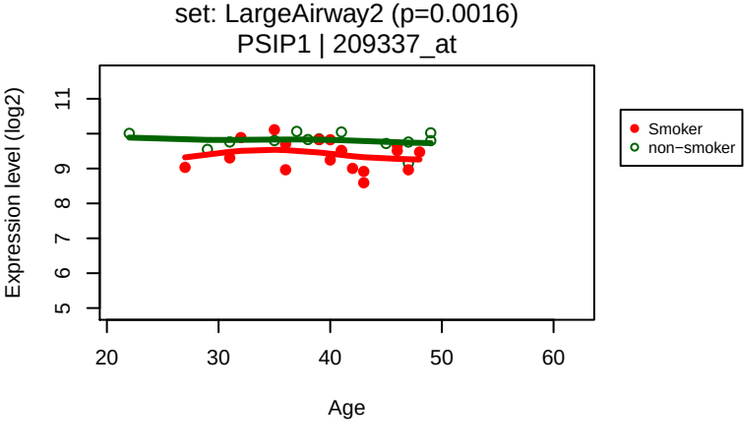

Gene expression ratio (smokers/nonsmokers)
gene: PSIP1 | 209337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3479
p=0.0193
p=0
p=0.0049
p=0.0016
p=0.0062
p=0.002
p=0.09
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

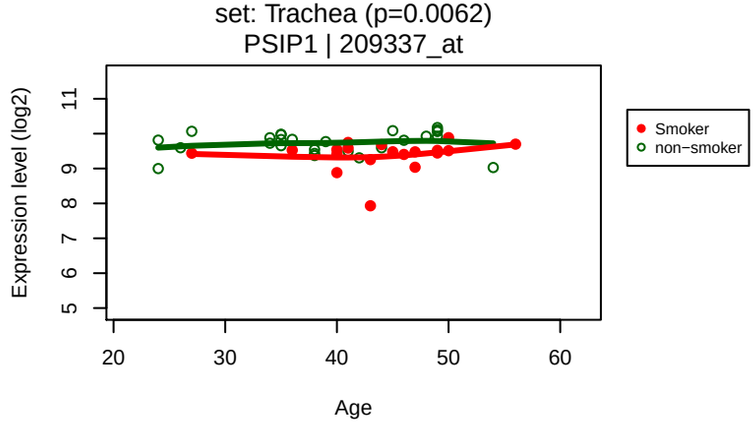

Gene expression ratio in Males
gene: PSIP1 | 209337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2331
p=0.0145
p=0.0295
p=0.1532
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSIP1 | 209337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2246
p=0.1252
p=0.042
p=0.3943
SmallAirway2
LargeAirway2
Trachea
Macrophage2

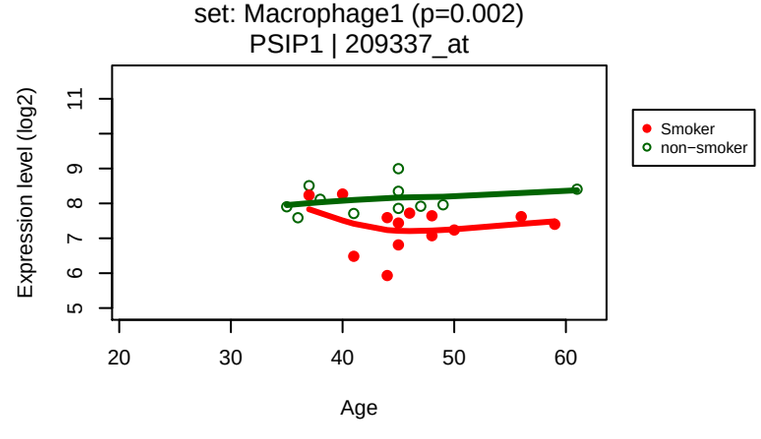

set: SmallAirway1 (p=0.3479)
PSIP1 | 209337_at
Expression level (log2)
Age
Smoker
non−smoker

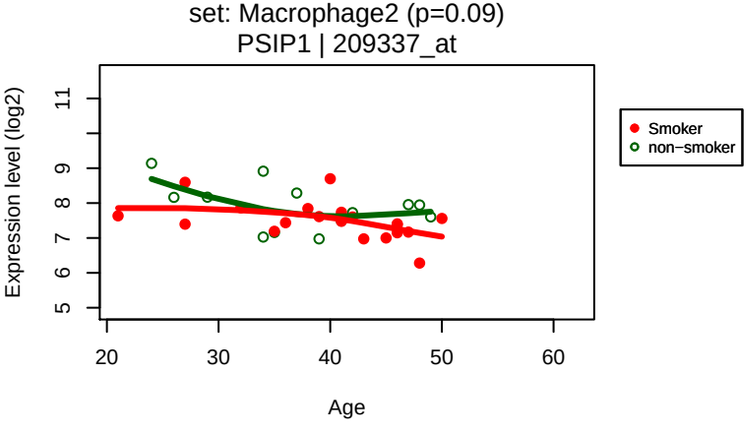

set: SmallAirway2 (p=0.0193)
PSIP1 | 209337_at
Expression level (log2)
Age
Smoker
non−smoker

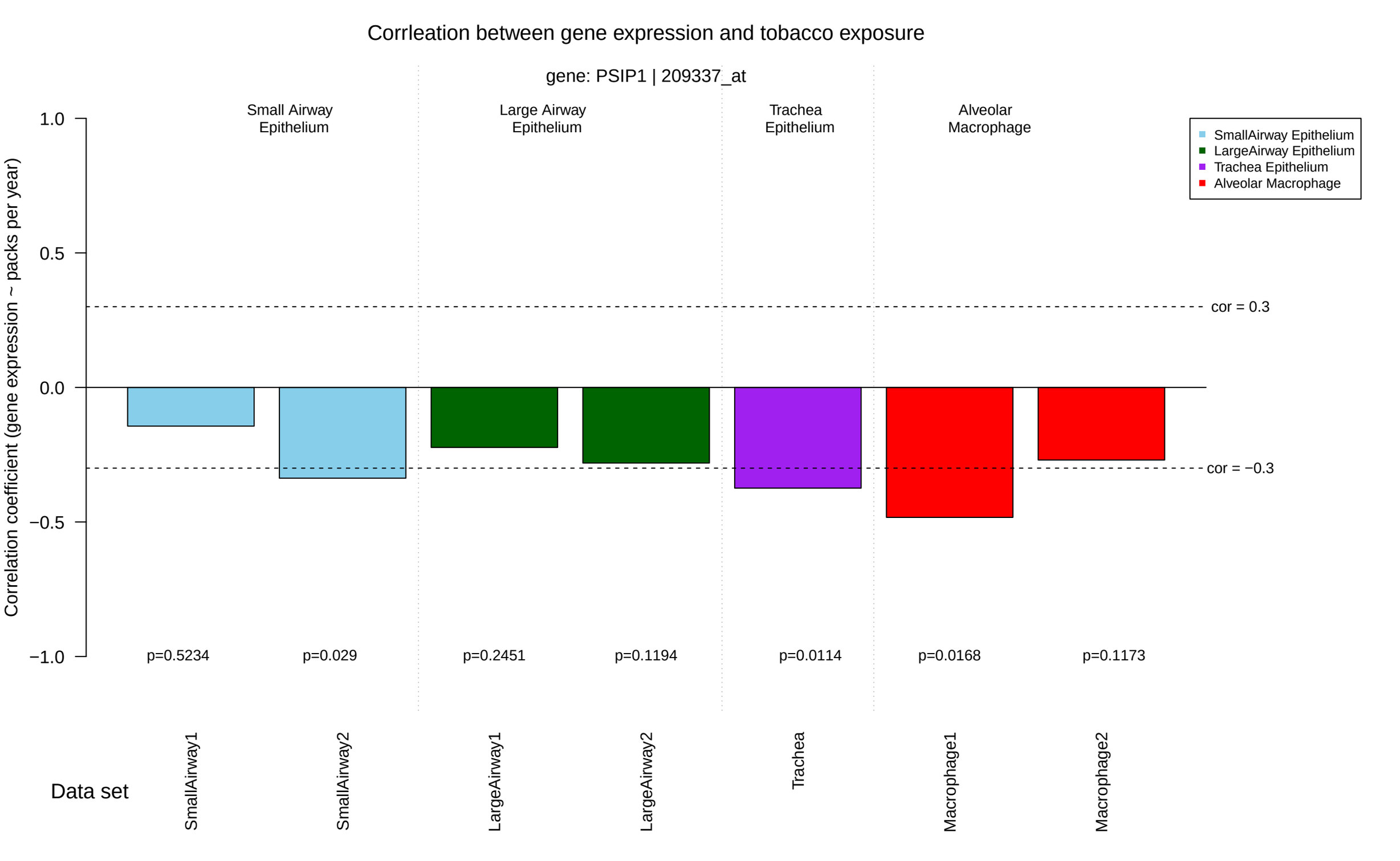

set: LargeAirway1 (p=0.0049)
PSIP1 | 209337_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0016)
PSIP1 | 209337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0062)
PSIP1 | 209337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.002)
PSIP1 | 209337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.09)
PSIP1 | 209337_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSIP1 | 209337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5234
p=0.029
p=0.2451
p=0.1194
p=0.0114
p=0.0168
p=0.1173
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2